

Recommended Next Phase

Trump Occurrence Forecasting Benchmark

What the locked test ruled out, what remains unproven, and the experiment that should come next.

Recommendation

Stop the current lexical adjacency-feature challenger. Preserve its **STOP** result as the valid conclusion of the completed sprint. Continue the broader research question only through a redesigned, target-conditioned neural experiment with rolling chronological validation and a newly sealed test period.

0.043319

Baseline B Brier

0.044434

Challenger Brier

+2.58%

relative error increase

**0.000259 to
0.002074**

bootstrap delta interval

What the sprint established

The final train-only adjacency feature path did not transfer reliably to the April to June 2026 locked test.
Timing and target-history information generalized better than the selected content features.
The current test set is exposed and cannot support further model selection.

What it did not establish

It did not test a trained target-conditioned transformer or LoRA model.
It did not show that all neural content representations are incapable of adding out-of-time signal.
It did not determine whether the failure came from representation weakness, temporal shift, or target construction.

Failure Diagnosis

The validation result was mildly positive, but the signal reversed on the locked chronological test. That pattern is more informative than a simple null result: it narrows the likely failure to **representation**, **temporal transfer**, or **target design**.



Figure 1: The content challenger beat Baseline B slightly on validation, then generalized worse on the locked test. Negative is favorable; positive is unfavorable.

Hypothesis	Evidence	How Phase 2 tests it
Representation failure	The final model selected five train-only adjacency features plus remaining duration. Configured semantic features were available but not selected.	Replace lexical adjacency with a jointly trained target-prefix encoder and discrete hazard head.
Temporal transfer failure	Train ended October 9, 2023; validation extended to April 7, 2026; the locked test covered April 12 to June 7, 2026. The validation gain did not survive this shift.	Use several rolling-origin validation windows and require consistent gains before opening a new test.
Target-design failure	The largest degradation was on high-frequency targets (Δ Brier = +0.009085); phrases were also worse (+0.001431).	Audit predictability by target family and redesign targets that are dominated by unconditional frequency rather than prefix evidence.

Interpretation. The evidence does not justify “more compute on the same model.” It justifies an experiment that can distinguish the three hypotheses above. Compute becomes useful only after the evaluation design can tell us why a model wins or fails.

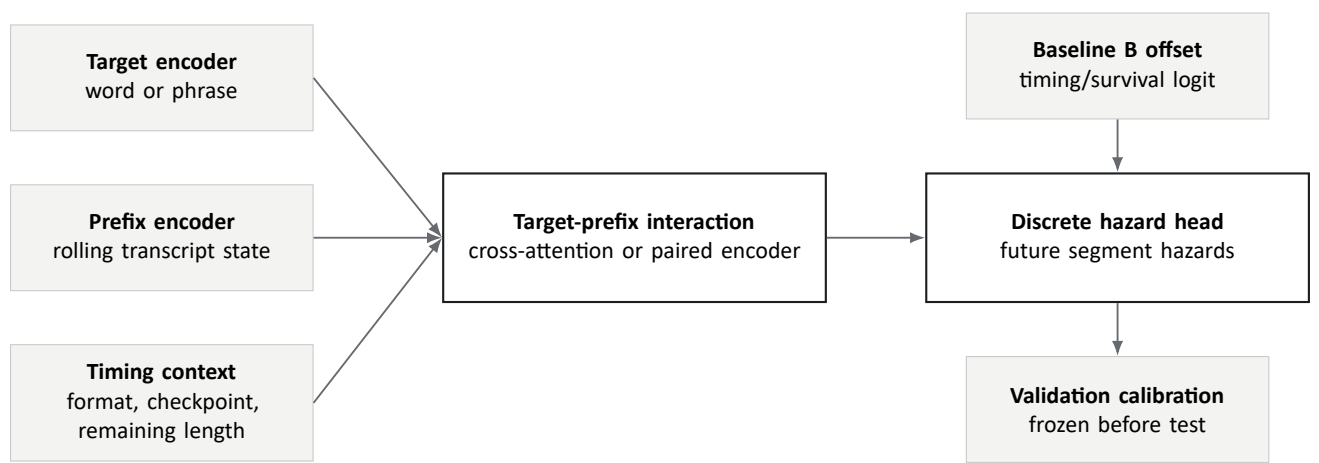
Recommended Phase 2 Architecture

Primary model: target-conditioned discrete-time neural hazard model

Given the target, transcript prefix, format, checkpoint, and expected remaining length, predict a hazard for the target’s first future occurrence in each remaining segment. Convert those hazards into the probability of occurrence before the transcript ends:

$$P(\text{occurs before end}) = 1 - \prod_{k=1}^K (1 - h_k) .$$

Baseline B remains a fixed logit offset. The measured delta therefore answers the original question directly: does neural content add information beyond timing and target history?



Candidate	Purpose	Role
Neural hazard model	Learns how target relevance evolves across the remaining transcript while preserving explicit time-to-event structure.	Primary model
Retrieval-augmented forecaster	Retrieves historically similar target-prefix states and uses their observed continuations as transparent evidence.	Diagnostic / ensemble
LoRA target-conditioned transformer	Tests whether parameter-efficient adaptation can learn richer target-prefix interactions than a frozen encoder.	GPU arm after Gate 1

Why this is materially different. The failed system summarized train-era lexical neighborhoods. The proposed system learns a shared interaction between an arbitrary target and the evolving transcript state, while modeling how opportunity decays as the transcript approaches its end.

Execution Sequence and Decision Gates

- Gate 0: rebuild the evaluation frontier.** Create multiple rolling chronological folds, preserve transcript-level separation, and reserve a genuinely new final test period or independent source corpus.
- Gate 1: prove content signal before expensive tuning.** Train a compact target-conditioned encoder and hazard head. Continue only if it improves Brier on most rolling folds, on $t > 0$ rows, and against a blank-prefix ablation.
- Gate 2: spend GPU on the strongest hypothesis.** Compare frozen encoder, retrieval augmentation, and LoRA under identical folds, calibration, target lists, and compute accounting. Select using validation only.
- Gate 3: freeze and open one new test.** Freeze architecture, hyperparameters, calibrator, and decision thresholds. Run all arms together once. Report transcript-level bootstrap intervals and contamination- pruned results.

Pre-test continuation gate	Minimum evidence	Why
Rolling chronological consistency	Positive Brier improvement on at least 3 of 4 folds; no large adverse fold	Prevents one favorable period from authorizing the test.
After-start usefulness	Positive improvement on $t > 0$ rows with transcript-bootstrap support	Confirms that live prefix content, not only $t = 0$, carries signal.
Content contribution	Full-prefix arm materially beats blank-prefix arm; content share at least 0.5	Demonstrates that transcript content is the source of improvement.
Target-family stability	No material regression on high-frequency targets; report phrases and cold targets	Addresses the largest failure concentration in the first sprint.
Calibration	Challenger ECE no more than 0.01 above Baseline B	Keeps probability quality aligned with the original objective.
Operational feasibility	Latency, memory, tokens, and dollar cost recorded for a complete pass	Prevents an accurate but unusable method from advancing.

Final recommendation

Do not reopen or tune against the current locked test. Preserve the completed **STOP** decision. Approve a separate Phase 2 only if its objective is to test a genuinely target-conditioned neural representation under rolling time-shift validation and a newly sealed evaluation.

Research grounding

The proposed design follows two established principles: dynamic time-to-event prediction should update risk as longitudinal context changes, and model selection under real-world distribution shift should evaluate performance across distinct environments rather than a single random split.

- [1] Changhee Lee, Jinsung Yoon, and Mihaela van der Schaar. *Dynamic-DeepHit: A Deep Learning Approach for Dynamic Survival Analysis With Competing Risks*. AAAI, 2020. AAAI article.
- [2] Pang Wei Koh et al. *WILDS: A Benchmark of in-the-Wild Distribution Shifts*. ICML, 2021. PMLR article.